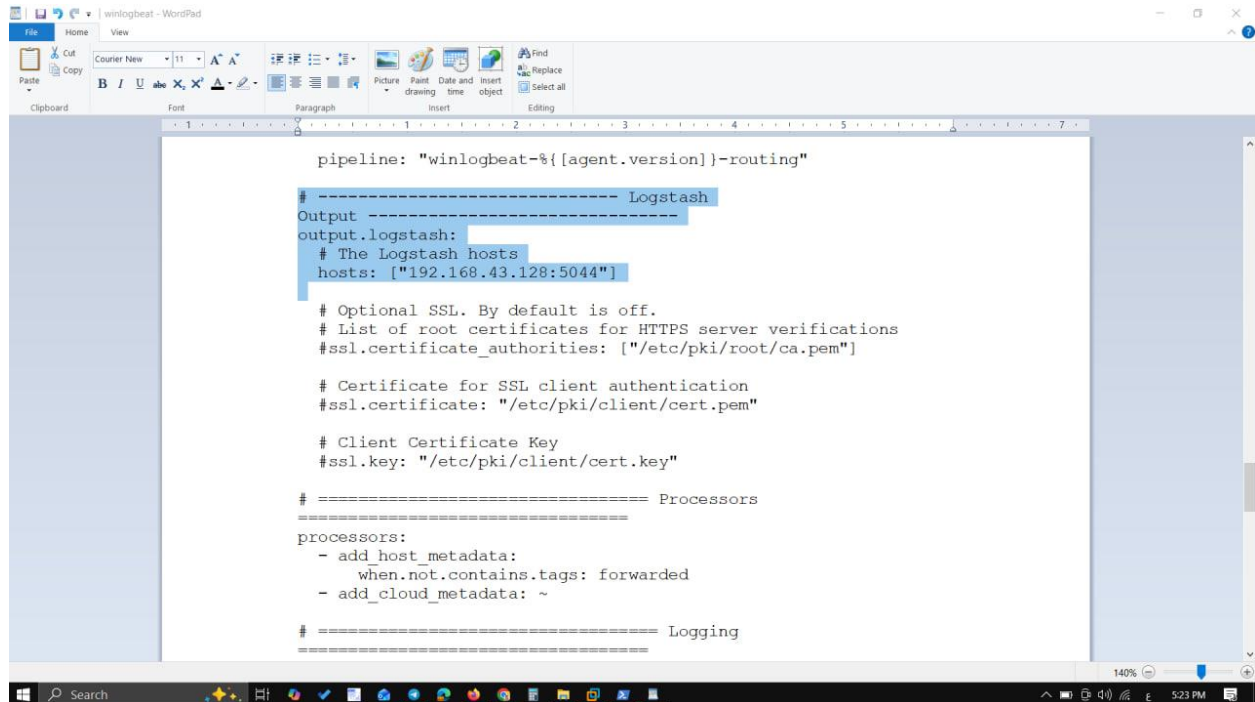


Install and Configure Logstash

(Winlogbeat to Logstash to Elasticsearch)

1. Configuring Winlogbeat Output to Logstash.

I will hash the elastic configuration because winlogbeat is single output either sends to Logstash or to ELK.



```
pipeline: "winlogbeat-%{[agent.version]}-routing"

# ----- Logstash
Output -----
output.logstash:
# The Logstash hosts
hosts: ["192.168.43.128:5044"]

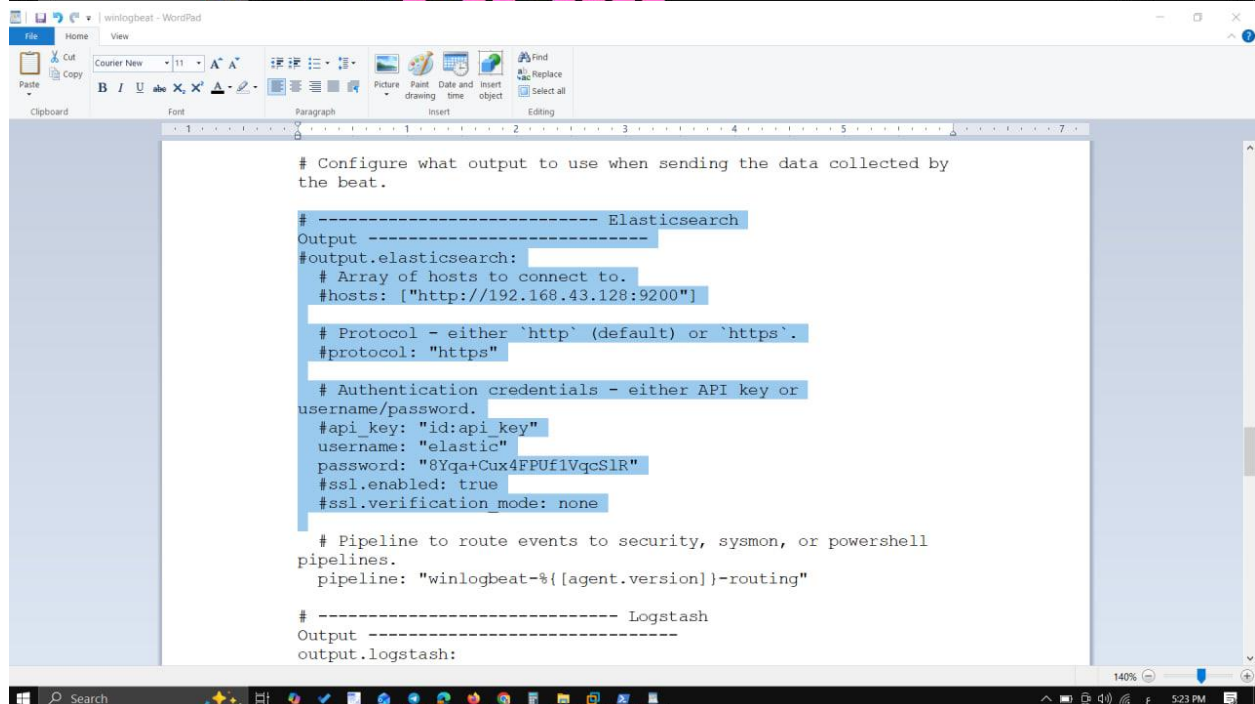
# Optional SSL. By default is off.
# List of root certificates for HTTPS server verifications
#ssl.certificate_authorities: ["/etc/pki/root/ca.pem"]

# Certificate for SSL client authentication
#ssl.certificate: "/etc/pki/client/cert.pem"

# Client Certificate Key
#ssl.key: "/etc/pki/client/cert.key"

# ===== Processors
processors:
- add_host_metadata:
  when.not.contains.tags: forwarded
- add_cloud_metadata: ~

# ===== Logging
```



```
# Configure what output to use when sending the data collected by
the beat.

# ----- Elasticsearch
Output -----
#output.elasticsearch:
# Array of hosts to connect to.
#hosts: ["http://192.168.43.128:9200"]

# Protocol - either `http` (default) or `https`.
#protocol: "https"

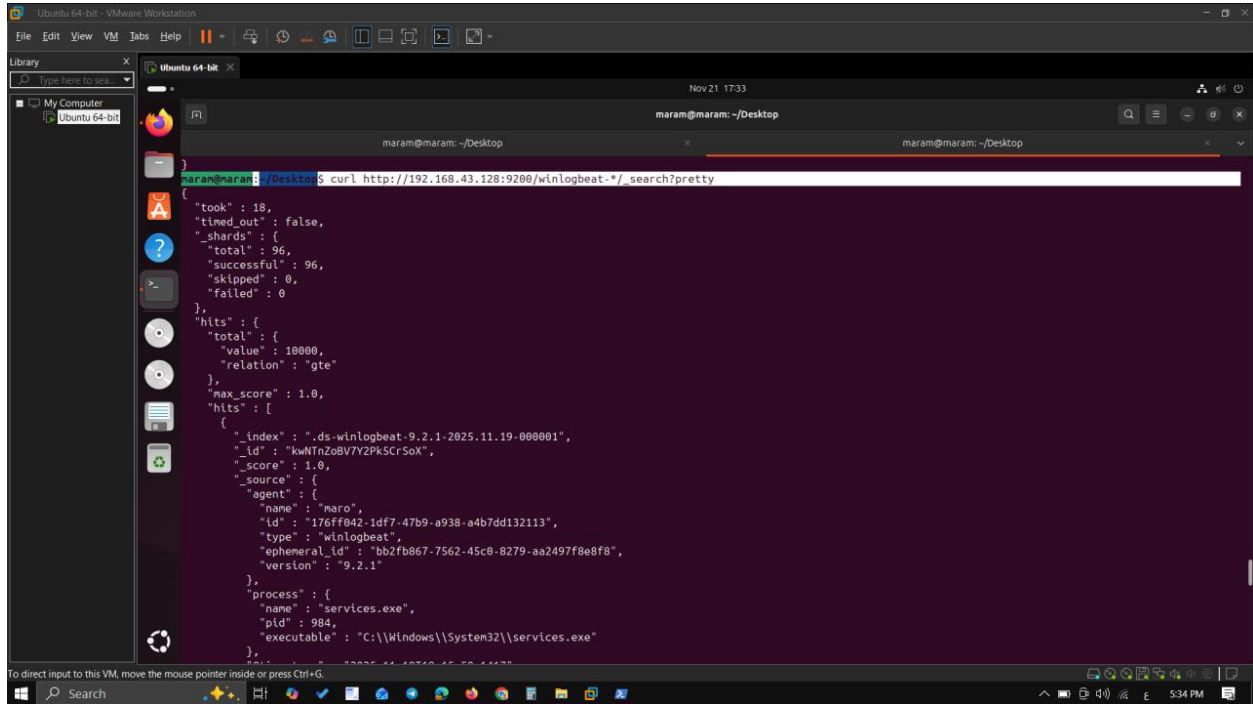
# Authentication credentials - either API key or
username/password.
#api_key: "id:api_key"
username: "elastic"
password: "8Yqa+Cux4FPuf1VqcSlR"
#ssl.enabled: true
#ssl.verification_mode: none

# Pipeline to route events to security, sysmon, or powershell
pipelines.
pipeline: "winlogbeat-%{[agent.version]}-routing"

# ----- Logstash
Output -----
output.logstash:
```

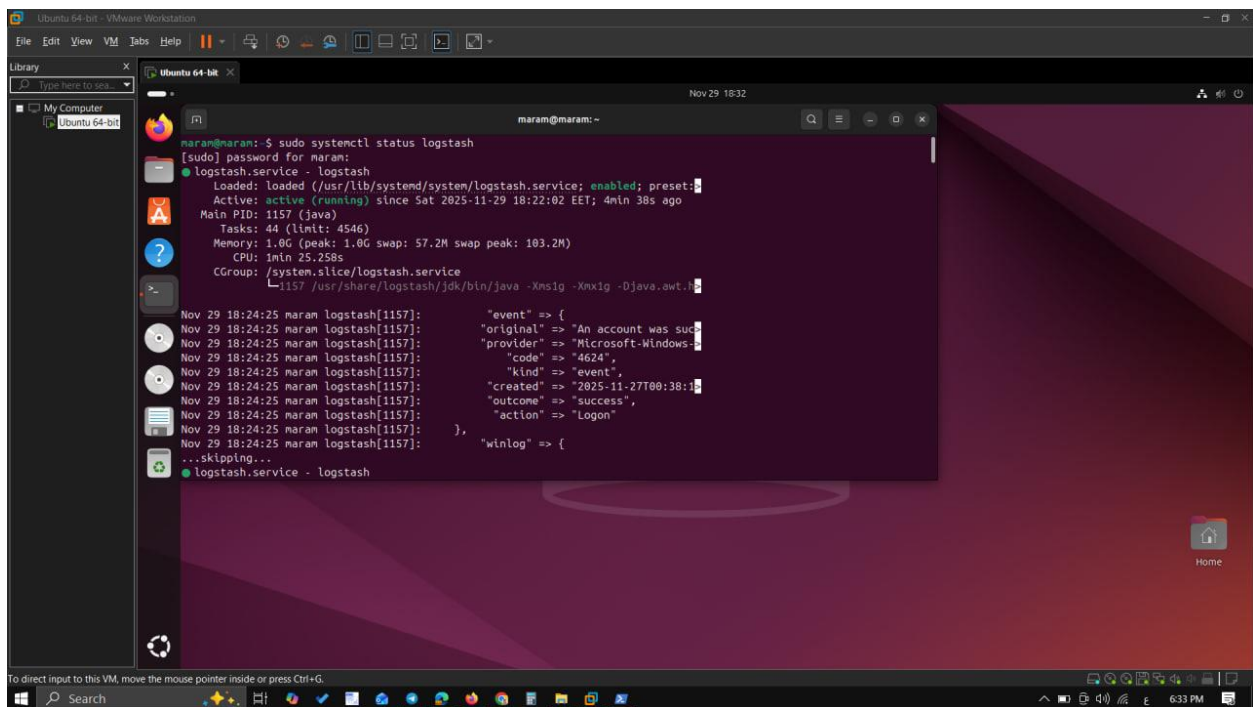
2.Verifying Winlogbeat Data in Elasticsearch

After Winlogbeat and Logstash are properly configured and running, you can verify that Windows event logs are being indexed into Elasticsearch using a curl command.



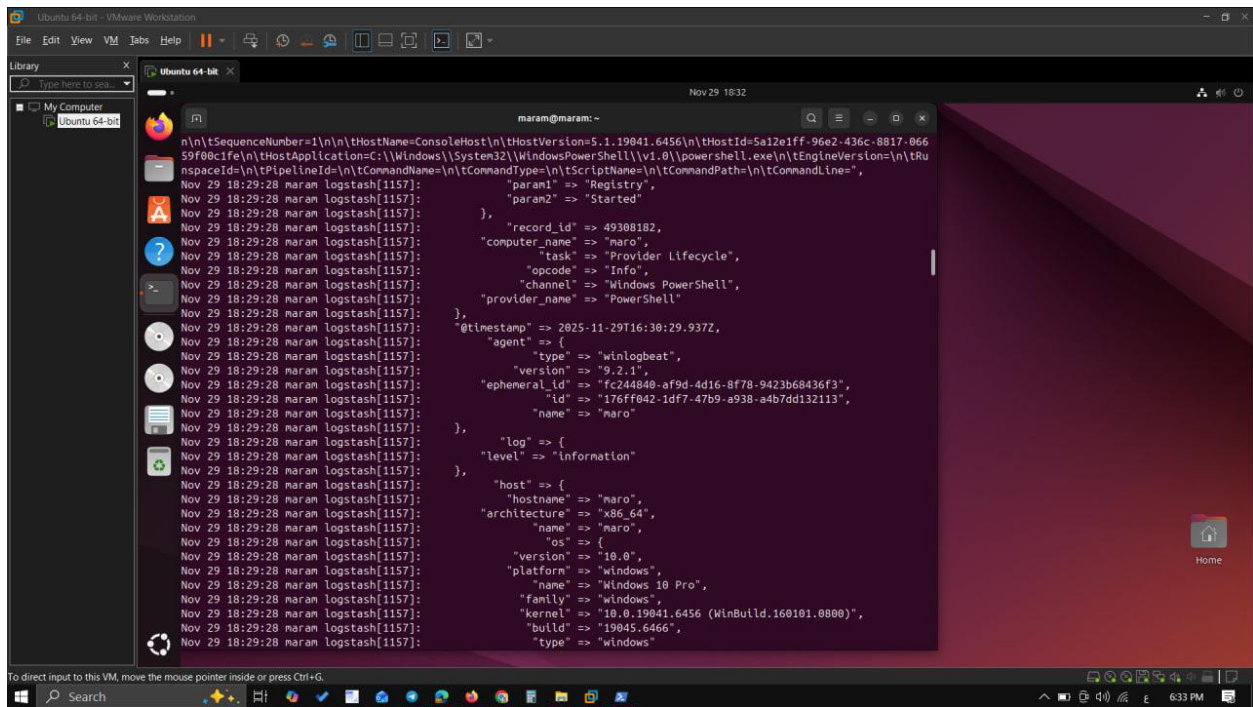
The screenshot shows a terminal window in a VM with the command `curl http://192.168.43.128:9200/winlogbeat-*/_search?pretty` and its output. The output is a JSON document representing a search result for Winlogbeat logs.

```
maram@maram: ~/Desktop
maram@maram:~/Desktop$ curl http://192.168.43.128:9200/winlogbeat-*/_search?pretty
{
  "took": 18,
  "timed_out": false,
  "_shards": {
    "total": 96,
    "successful": 96,
    "skipped": 0,
    "failed": 0
  },
  "hits": {
    "total": {
      "value": 10000,
      "relation": "gte"
    },
    "max_score": 1.0,
    "hits": [
      {
        "_index": ".ds-winlogbeat-9.2.1-2025.11.19-000001",
        "_id": "kWTnZobV7Y2Pk5Cr5oX",
        "_score": 1.0,
        "_source": {
          "agent": {
            "name": "maram",
            "id": "176ff042-1df7-47b9-a938-a4b7dd132113",
            "type": "winlogbeat",
            "ephemeral_id": "bb2fb867-7562-45c8-8279-aa2497f8e8f8",
            "version": "9.2.1"
          },
          "process": {
            "name": "services.exe",
            "pid": 984,
            "executable": "C:\\Windows\\System32\\services.exe"
          }
        }
      }
    ]
  }
}
```



The screenshot shows a terminal window in a VM with the command `sudo systemctl status logstash` and its output. The output shows that logstash is active and running. Below the status output, there are several log entries from logstash showing a successful login event.

```
maram@maram: ~
maram@maram:~$ sudo systemctl status logstash
[sudo] password for maram:
logstash.service - logstash
Loaded: loaded (/usr/lib/systemd/system/logstash.service; enabled; preset:
Active: active (running) since Sat 2025-11-29 18:22:02 EET; 4min 38s ago
Main PID: 1157 (java)
Tasks: 44 (limit: 4546)
Memory: 1.0G (peak: 1.0G swap: 57.2M swap peak: 103.2M)
CPU: 1min 25.258s
CGroup: /system.slice/logstash.service
└─1157 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -Djava.awt.headless=
Nov 29 18:24:25 maram logstash[1157]: "event" => {
Nov 29 18:24:25 maram logstash[1157]: "original" => "An account was suc
Nov 29 18:24:25 maram logstash[1157]: "provider" => "Microsoft-Windows-
Nov 29 18:24:25 maram logstash[1157]: "code" => "4624",
Nov 29 18:24:25 maram logstash[1157]: "kind" => "event",
Nov 29 18:24:25 maram logstash[1157]: "created" => "2025-11-27T00:38:1
Nov 29 18:24:25 maram logstash[1157]: "outcome" => "success",
Nov 29 18:24:25 maram logstash[1157]: "action" => "Logon"
Nov 29 18:24:25 maram logstash[1157]: },
Nov 29 18:24:25 maram logstash[1157]: "winlog" => {
...skipping...
logstash.service - logstash
```



Complete Logstash Pipeline Summary

Winlogbeat (Windows) → Logstash (Port 5044) → Elasticsearch (Port 9200) → Kibana (Port 5601)

Benefits of using Logstash between Winlogbeat and Elasticsearch:

- Centralized data processing and transformation
- Multiple input sources aggregation
- Data filtering and enrichment before indexing
- SSL/TLS termination point
- Buffer management during Elasticsearch downtime